

Ivonne Mariela García Martínez, Ph.D.

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Education

- 2016 – 2020 📖 **Ph.D. Atmospheric and Environmental Sciences.** The University of Edinburgh. Edinburgh, UK.
Thesis title: *Mechanisms of climate variation in the broader Mexican-United States region: the role of atmospheric circulation and anthropogenic forcing.*
- 2012 – 2015 📖 **M.Sc. Physical Oceanography.** Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, Mexico.
Thesis title: *Ocean-atmosphere variability related with the midsummer drought in the CFSR reanalysis.*
- 2007 – 2011 📖 **B.Sc. Atmospheric Sciences.** Universidad Veracruzana. Xalapa, Mexico.
Graduated by high GPA.

Research Publications

Peer-reviewed articles

- 1 C. Raymond, **I. M. García-Martínez**, C. Rogers, *et al.*, “Compound weather and climate events in 2025,” *Nature Reviews Earth & Environment*, 2026, accepted for publication.
- 2 **I. M. García-Martínez**, J. Sheinbaum, L. Zavala Sansón, J. Jouanno, and E. Pallàs-Sanz, “Upper-ocean response to strong wind forcing in a Loop Current Eddy,” *Journal of Physical Oceanography*, vol. 55, no. 12, pp. 2401–2420, 2025. 🔗 DOI: 10.1175/JPO-D-25-0022.1.
- 3 L. Zavala Sansón, **I. M. García-Martínez**, and J. Sheinbaum, “Power input generated by a linear wind field on circular and elliptical ocean vortices,” *Journal of Physical Oceanography*, vol. 55, no. 9, pp. 1493–1506, 2025. 🔗 DOI: 10.1175/JPO-D-24-0114.1.
- 4 L. Zavala Sansón, **I. M. García-Martínez**, and J. Sheinbaum Pardo, “Nonlinear surface Ekman effects on cyclonic and anticyclonic vortices,” *Journal of Fluid Mechanics*, vol. 971, A35, 2023. 🔗 DOI: 10.1017/jfm.2023.688.
- 5 **I. M. García-Martínez** and M. A. Bolasina, “Identifying the evolving human imprint on heat wave trends over the United States and Mexico,” *Environmental Research Letters*, vol. 16, no. 9, p. 094 039, 2021. 🔗 DOI: 10.1088/1748-9326/ac1edb.
- 6 **I. M. García-Martínez**, M. A. Bolasina, and S. Undorf, “Strong large-scale climate response to North American sulphate aerosols in CESM,” *Environmental Research Letters*, vol. 15, no. 11, p. 114 051, 2020. 🔗 DOI: 10.1088/1748-9326/abbe45.
- 7 **I. M. García-Martínez** and M. A. Bolasina, “Sub-monthly evolution of the Caribbean Low-Level Jet and its relationship with regional precipitation and atmospheric circulation,” *Climate Dynamics*, vol. 54, pp. 4423–4440, 2020. 🔗 DOI: 10.1007/s00382-020-05237-y.

Book chapters

- 1 A. Tejeda Martínez, D. R. Fitzjarrald, M. Pérez Méndez, *et al.*, “Ciclos diurnos y circulación local (Diurnal cycles and local circulation),” in *Un laboratorio natural del clima: la interacción costa-montaña en el centro del Golfo de México*, A. Tejeda Martínez, M. Pérez Méndez, and I. R. Méndez Pérez, Eds., Xalapa, México: Universidad Veracruzana, 2020, pp. 46–91.
- 2 A. Tejeda Martínez, **I. M. Garcia-Martínez**, G. Gómez-Azpeitia, I. R. Méndez-Pérez, and P. J. González Viveros, “Procedimientos simples para la estimación de radiación Solar en México (Simple procedures for the estimation of solar radiation in Mexico),” in *Prontuario solar de México*, G. Gómez-Azpeitia and A. Tejeda Martínez, Eds., Colima, México: Universidad de Colima and Universidad Veracruzana, 2015, pp. 103–112.
- 3 A. Tejeda Martínez, **I. M. Garcia-Martínez**, I. R. Méndez-Pérez, U. Miranda-Miranda, and J. V. López-Méndez, “Radiación solar en México (Solar radiation in Mexico),” in *Prontuario solar de México*, G. Gómez-Azpeitia and A. Tejeda Martínez, Eds., Colima, México: Universidad de Colima and Universidad Veracruzana, 2015, pp. 51–68.

Manuscripts under review and in preparation

- 1 **I. M. Garcia-Martínez** and M. A. Bollasina, “Escalating risks of compound heat wave-drought events in Mexico and the United States: future projections and population exposure,” In preparation. To be submitted to *Earth’s Future*.
- 2 **I. M. Garcia-Martínez**, J. Jouanno, J. Rodríguez, *et al.*, “Dynamical response of a Loop Current Eddy to strong wind forcing in high-resolution coupled ocean-atmosphere simulations,” In preparation. To be submitted to *Journal of Geophysical Research: Oceans*.
- 3 J. L. Rodríguez Solís, **I. M. Garcia-Martínez**, and S. J., “From prediction to process understanding: the rapid intensification of hurricane Otis in coupled ocean-atmosphere simulations,” In preparation.
- 4 C. Romero González, P. Cetina-Heredia, **I. M. Garcia-Martínez**, and J. Benthuyzen, “Below the Surface: tracing the depth footprint of Marine Heatwaves in the Caribbean,” To be submitted to *Ocean Science*.
- 5 C. Zúñiga Villareal, T. Cavazos, **I. M. Garcia-Martínez**, C. Raymond, and S. Gershunov, “Unraveling the synoptic drivers of heatwaves in Northern Mexico and the southwestern United States,” To be submitted to *Climate Dynamics*.

Manuscript reviews

- 📖 Climate Dynamics
- 📖 International Journal of Climatology
- 📖 Journal of Physical Oceanography
- 📖 GEOS

Awarded scholarships

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| 2016 – 2020 | <ul style="list-style-type: none"> 📖 PhD Scholarship from the National Council for Science and Technology (CONA-CyT, Mexico). 📖 Complementary PhD scholarship from the University of Edinburgh, UK. |
| 2012 – 2014 | <ul style="list-style-type: none"> 📖 MSc Scholarship from the National Council for Science and Technology (CONA-CyT, Mexico). |

Academic employment

- 2024 –  **Research Associate, Department of Physical Oceanography.** Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, México.
- 2021 – 2023  **Postdoctoral Researcher, Department of Physical Oceanography.** Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, México.
- 2015 – 2016  **Junior Researcher (M.Sc. level), Department of Physical Oceanography.** Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, México.
- 2011 – 2012  **Research Assistant, Applied Climatology Group.** Universidad Veracruzana. Xalapa, México.

Research visits

- 2022 –  **Remote visiting researcher, School of GeoSciences.** The University of Edinburgh. Edinburgh, UK.
- 2025  **Laboratoire d'Etudes en Géophysique et Oceanographie Spatiales, LEGOS.** Toulouse, France.

Teaching experience

- 2022, 2025 – 2026  **Climatology and Climate Change.** Postgraduate program in Physical Oceanography. Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, México. Lecturer.
- 2026  **Research Seminar.** Postgraduate program in Physical Oceanography. Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, México. Coordinator.
- 2016 – 2018  **Earth Modelling and Prediction.** School of GeoSciences. The University of Edinburgh. Edinburgh, UK. Teaching assistant.

Invited talks

- 2022  **Weather of 2021 in the context of the long-term climate change.** "Understanding the Weather of 2021" virtual event of the Royal Meteorological Society of the UK. 26 February 2022. <https://www.rmets.org/event/virtual-understanding-weather-2021>
-  **State of the climate in 2021 under the context of climate change.** Universidad Veracruzana. Invited talk as part of the University's celebrations on the World Meteorological Day. 23 March 2022.
- 2021  **Heat wave trends in Mexico and the United States: contribution of anthropogenic aerosols, greenhouse gases and internal variability.** Universidad Veracruzana. Invited seminar in the 45th Anniversary of the School of Atmospheric Sciences. 6 May 2021.

Related professional and technical skills

- Languages  Strong reading, writing and speaking skills in English. Native Spanish and basic Portuguese.

Related professional and technical skills (continued)

Coding	Python, Matlab, shell, NCO, CDO, GraDs, and high performance computing
Modelling	WRF (atmospheric model), NEMO and CROCO (ocean models), OASIS (coupler)
Version control	Git
Editing	L ^A T _E X

Additional training

- 2018  **ICTP Summer School** on Theory, Mechanisms and Hierarchical Modelling of Climate Dynamics: Multiple Equilibria in the Climate System. Trieste, Italy, 25-30 June 2018. At International Centre for Theoretical Physics.
-  **WCRP Grand Challenge** on Clouds, Circulation and Climate Sensitivity: 2nd meeting on Monsoons and Tropical Rain Belts. Trieste, Italy, 2 - 5 July 2018. At International Centre for Theoretical Physics.
- 2017  **GOTHAM Summer School** on Global Teleconnections in the Earth's Climate System: Processes, Modelling and Advanced Analysis Methods. Potsdam, Germany, 18-22 September 2017. At Potsdam Institute for Climate Impact Research.
- 2014  **XXII Alpine Summer School** on Dynamics, Stochastics and Predictability of the Climate System. Valsavarenche, Valle d'Aosta, Italy, June 9-18, 2014.

Participation in conferences and other academic activities

-  Ocean Sciences Meeting (**OS**). Glasgow, UK, 23-27 February 2026. An overview of marine heat waves in the Caribbean Sea; their characteristics, drivers and potential impacts on coral reefs. Cetina P., Allende-Arandía, M. E., **García-Martínez, I. M.**, Arévalo, X, Romero, C., and Gómez, M. eLightning presentation.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 26 October to November 1, 2025. Observational diagnosis of the footprint depth of marine heat waves in the Caribbean Sea. Romero, C., Cetina P and **García-Martínez, I. M.** Oral presentation.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 26 October to November 1, 2025. Modulation of vertical velocity in an anticyclonic vortex due to the passage of Nortes in the Gulf of Mexico. Jiménez, K., **García-Martínez, I. M.** and Others. Poster.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 27 October to November 2, 2024. Compound hot and dry events in Mexico: current scenarios and projections to 2050 based on models from the HIGHRESMIP project. **García-Martínez, I. M.** and M. Bollasina. Oral presentation.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 27 October to November 2, 2024. Coupled and non-coupled numerical simulations of the evolution of a Loop Current Eddy during a Norte (strong-wind) event. J. Sheinbaum, **García-Martínez, I. M.**, J. Jouanno and Others. Oral presentation.

Participation in conferences and other academic activities (continued)

- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 29 October to November 3, 2023. Towards a coupled ocean-atmosphere modelling system of the Gulf of Mexico: eddy-wind interactions. **García-Martínez, I. M.** and J. Sheinbaum. Oral presentation.
- European Geosciences Union (**EGU**) General Assembly. Vienna, Austria, April 23-28, 2023. Current feedback in an anticyclonic eddy during the passage of cold frontal systems over the Gulf of Mexico. **García-Martínez I. M.** and J. Sheinbaum. Poster.
- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 30 October to November 4, 2022. Influence of strong winds in an anticyclonic eddy in the Gulf of Mexico. **García-Martínez, I. M.** and J. Sheinbaum. Oral presentation.
- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 30 October to November 4, 2022. Heat wave trends in Mexico and the United States: contribution from anthropogenic aerosols, greenhouse gases and internal variability. **García-Martínez, I. M.** and M. Bolasina. Oral presentation.
- World Climate Research Program (**WCRP**) virtual workshop on attribution of multiannual to decadal changes in the climate system. September 22-24, 2021. Identifying the evolving human imprint on heat wave trends over Mexico and the United States. **García-Martínez, I. M.** and M. Bolasina. Oral presentation.
- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, November 2-8, 2019. Sub-monthly evolution of the Caribbean low-level jet and its relationship with precipitation in Mexico. **García-Martínez, I. M.** and M. Bolasina. Oral presentation.
- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, November 2-8, 2019. Anthropogenic aerosols as drivers of Mexican and Central American climate. **García-Martínez, I. M.** and M. Bolasina. Poster.
- European Geosciences Union (**EGU**) General Assembly. Vienna, Austria, 7-12 April 2019. Anthropogenic aerosols as drivers of Mexican and Central American climate. **García-Martínez I. M.** and M. Bolasina. Poster.
- European Geosciences Union (**EGU**) General Assembly. Vienna, Austria, April 8-13, 2018. Sub-monthly variability of the Caribbean Low-Level Jet and its relationship to precipitation and atmospheric circulation in Mexico and Central America. **García-Martínez I. M.** and M. Bolasina. Poster.
- Virtual workshop of the Climate Variability and Predictability Program (**US CLIVAR**): Observing and Modelling Climate Variability in the Intra-Americas Seas and Impacts on the Continental Americas and the Caribbean. September 9-11, 2015. Ocean-atmosphere variability related with the midsummer drought in the CFSR reanalysis. **García-Martínez, I. M.** and J. Sheinbaum. Poster.
- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, November 2-8, 2014. Ocean-atmosphere variability related with the midsummer drought in the CFSR reanalysis. **García-Martínez, I. M.** and J. Sheinbaum. Oral presentation.

Participation in conferences and other academic activities (continued)

- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, October 28-November 2, 2012. Empirical model for the estimation of the hourly diffuse radiation during clear-sky conditions in Xalapa city. López-Velázquez, E. and **García-Martínez I. M.** Poster.
- **XVI Congresso Brasileiro de Meteorologia**. Belém do Pará, Brazil. September 13-17, 2010. Analysis of the minimum temperature in the 2009-2010 Winter season for the Central Mexico region. **García-Martínez, I. M.**, Gómez, G., González, P., Jofre, R., Muñoz C., and Nava, R. Oral presentation.